

Reproduction of LAION-AI/CLIP_benchmark: zero-shot classification of OpenCLIP ViT-B-32 (LAION-2B) on CIFAR-10, CIFAR-100 and STL-10

K-Veritas Team

Submitted by K-Veritas Team

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Independent reproduction · Artifact evaluation · Computer vision

Abstract

Ran the clip_benchmark CLI from the pinned commit (installed editable) to evaluate the OpenCLIP ViT-B-32 model with the laion2b_s34b_b79k weights in zero-shot classification on the CIFAR-10, CIFAR-100 and STL-10 test sets, with the default English class names and prompt templates, batch size and AMP settings. Top-1, top-5 and mean per-class recall are read from the result JSON files the tool wrote during the sealed run. Datasets downloaded by torchvision and weights fetched from Hugging Face before the sealed run. Single NVIDIA RTX 5060 Ti 16 GB, PyTorch 2.11 cu128, Python 3.12.

About this record

Title, authors and abstract were provided by the authors and reviewed for completeness, not for scientific soundness. Every number in this record is copied from the signed data of the sealed reports [2], which remain authoritative.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	OpenCLIP ViT-B-32 zero-shot	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: OpenCLIP ViT-B-32 zero-shot

Run	Metric	Final value	Step
1	cifar10_acc1	0.9358	
1	cifar10_acc5	0.9981	
1	cifar10_mean_per_class_recall	0.9358	
1	cifar100_acc1	0.7555	
1	cifar100_acc5	0.9386	
1	cifar100_mean_per_class_recall	0.7551	
1	stl10_acc1	0.965625	
1	stl10_acc5	0.999375	
1	stl10_mean_per_class_recall	0.965875	

Run	Inline claim	Value	Phase
1	cifar10_acc1	0.9358	
1	cifar10_acc5	0.9981	
1	cifar100_acc1	0.7555	
1	cifar100_acc5	0.9386	
1	stl10_acc1	0.965625	
1	stl10_acc5	0.999375	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	45.2 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	f1772a35351677bae12ec8889312a056e253f363f56fb52bc41c816578d37285	481 KB
r1.kvbundle.zip	5d1e9b98d65e4e03f6ecbe6f288746a27db085ef7e7ea942a730e95b36157e9f	1.9 MB

5 Links and licence

Paper: <https://arxiv.org/abs/2212.07143>

Reproduction of: https://github.com/LAION-AI/CLIP_benchmark/tree/ee14a9a8a9bcde09d51f4a4b1dff5a75182a3596

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: clip, zero-shot, openclip, cifar-10, cifar-100, stl-10.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (OpenCLIP ViT-B-32 zero-shot), kv:2610.00014v1/r1. Data hash cc70399aa96216cc54eafde7a82fa4a7b98ee14b581fa7ae5559bd639928281c.
<https://kveritas.org/records/2610.00014v1/r1.pdf>